

Exact projection links for one approximately inner automorphism

MathlibAnnex.CStarAlgebra.exists_shell_family_of_approximately_inner

theorem

A point-norm approximation followed by close-projection conjugacy gives exact support identities for an arbitrary projection family.

Statement

Let A be a nonzero unital complex C^* -algebra with its usual positive order, and let $\alpha : A \rightarrow A$ be a complex star automorphism. Suppose that for every finite $F \subset A$ and $\varepsilon > 0$ there is $g \in U(A)$ satisfying $\|\alpha(a) - gag^*\| < \varepsilon$ for all $a \in F$. For any set I and family of self-adjoint projections $(q_i)_{i \in I}$ in A , there are elements $(w_i)_{i \in I}$ such that $w_i^*w_i = \alpha(q_i)$ and $w_iw_i^* = q_i$ for every i .

Assumptions

The same automorphism α and its finite-set approximation property are fixed for the whole family. The index set is arbitrary. The projections need not be nonzero, orthogonal, or countably indexed, and no condition on their sum is imposed.

Conclusion

Each w_i is a partial-isometry link with initial support $\alpha(q_i)$ and final support q_i . The resulting links all use the one specified α , although the auxiliary approximating and correcting unitaries may depend on i .

The result gives algebraic support equations for the family. It asserts no convergence of a sum of links, no common correcting unitary, and no uniqueness of the chosen links. No pure-state or homogeneity hypothesis is needed once α and its approximate innerness are supplied.

Proof route

For one projection q , approximate $\alpha(q)$ within distance 1 by ggg^* . Close self-adjoint projections are exactly unitarily conjugate, so choose t with $t\alpha(q)t^* = ggg^*$. Then $w = g^*t\alpha(q)$ has the required supports. Apply this argument separately to each index and choose one link at each index.

▼ Proof steps

1. Approximate on the singleton $\{q\}$ with tolerance 1. Both $e = \alpha(q)$ and ggg^* are self-adjoint projections and $\|e - ggg^*\| < 1$. The close-projection conjugacy theorem supplies $t \in U(A)$ with $tet^* = ggg^*$.
2. For $w = g^*te$, the unitary identities and $e = e^* = e^2$ give $w^*w = et^*gg^*te = e$. The other support is $ww^* = g^*tet^*g = q$. Thus the direction of the two support equations is fixed.

3. Use choice on the individual existence statements for q_i . This produces a function $i \mapsto w_i$ with both equations at every index, including the vacuous case of an empty index set.

Main citations

- The stated existence or structural result — `MathlibAnnex.CStarAlgebra.exists_shell_family_of_approximately_inner`
- Single-projection construction at tolerance one — `MathlibAnnex.CStarAlgebra.exists_shell_of_approximately_inner`
- Close-projection unitary conjugacy — `IsStarProjection.exists_unitary_conjugate_of_norm_sub_lt_one`
- Exact link formula — `MathlibAnnex.CStarAlgebra.matchedLink`
- Both support identities for the link — `MathlibAnnex.CStarAlgebra.matchedLink_supports`

Supporting route explanation

Lean source signature (exact)

```
theorem exists_shell_family_of_approximately_inner
  (A : Type u) {ι : Type v}
  [CStarAlgebra A] [PartialOrder A] [StarOrderedRing A] [Nontrivial A]
  (alpha : A ≈*a[C] A)
  (happrox : ∀ (F : Finset A) (epsilon : ℝ), 0 < epsilon →
    ∃ g : unitary A, ∀ a ∈ F,
      ‖alpha a - (g : A) * a * star (g : A)‖ < epsilon)
  (f : ι → A) (hf : ∀ i, IsStarProjection (f i)) :
  ∃ w : ι → A, ∀ i,
    star (w i) * w i = alpha (f i) ∧ w i * star (w i) = f i
```

In the source	Mathematical meaning
<code>A; [CStarAlgebra A]; [PartialOrder A]; [StarOrderedRing A]; [Nontrivial A]</code>	A nonzero unital complex C*-algebra A with its usual positive order. This theorem is not restricted to CAR.
<code>ι : Type v; alpha : A ≈*_a[C] A</code>	An arbitrary index set I and one fixed complex star automorphism $\alpha : A \rightarrow A$.
<code>happrox : ∀ (F : Finset A) (epsilon : ℝ), 0 < epsilon → ...</code>	For every finite $F \subset A$ and $\varepsilon > 0$, there is one unitary $g \in A$ satisfying $\ \alpha(a) - gag^*\ < \varepsilon$ for all $a \in F$. This assumption applies to the same α for the whole family.
<code>f : ι → A; hf : ∀ i, IsStarProjection (f i)</code>	The source family $f(i)$ is the family $q_i \in A$ in the text; each $q_i = q_i^* = q_i^2$. The index set need not be countable and the projections need not be orthogonal or nonzero.
<code>∃ w : ι → A, ∀ i</code>	There is one family $i \mapsto w_i$ of elements of A with both following support equations for every $i \in I$.
<code>star (w i) * w i = alpha (f i)</code>	The initial support is $w_i^* w_i = \alpha(q_i)$.

In the source	Mathematical meaning
<code>w i * star (w i) = f i</code>	The final support is $w_i w_i^* = q_i$. Both equations concern the same w_i and the fixed α ; these links need not be unitaries.

Further source notes: In the single-projection proof `matchedLink g t (alpha f)` is exactly $g^* t \alpha(q)$. The first support is `star w * w`; it must not be interchanged with the second.

Earlier published Card and PDF

▼ Exact content identity

Declaration: `MathlibAnnex.CStarAlgebra.exists_shell_family_of_approximately_inner`

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Accepted source guide SHA-256: `a230e2e80d7461106b2ebd36e31ab8a848be8cef9db391c301e39f6b46751bcf`

Source commit: `437e6e46228bbb8e91211ebded349d7a30020e73`